

NAGA JAHNAVI CHAKKA

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SUMMARY

Data Analyst proficient in Python, SQL, Power BI, and Tableau. Experienced with large datasets and delivering actionable insights to inform business decisions. Skilled in deploying machine learning models in cloud platforms. Adaptable and collaborative, excelling in both independent and team-oriented environments to drive impactful data solutions.

EDUCATION

Binghamton University, State University of New York, Thomas J. Watson College of Engineering and Applied Science

Master of Science in Computer Science

Aug 2023 – Dec 2024

Cumulative GPA: 3.7/4.00

Vellore Institute of Technology, VITAP University

AP, India

Bachelor of Technology Computer Science

Aug 2019 – May 2023

Cumulative GPA: 3.8/4.00

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, PHP, HTML/CSS

Software and OS: Alteryx, PowerBI, Tableau, Jupyter Notebook, Git (Version Control), Linux, Pytorch, Maven Excel, Airflow

Databases: MySQL, Postgre SQL, SQLite, MongoDB, Redis, Cassandra

Libraries/Frameworks: Django, TensorFlow, Keras, NumPy, Matplotlib, Flask, Pandas, Sklearn

Tools: AWS, Azure, Snowflake, Jira, Docker, Kubernetes, Git/GitHub, Apache Airflow, Power BI, Tableau, MLflow, Jenkins

PROFESSIONAL EXPERIENCE

The Graduate Student Assistant, Project Assistant | New York, USA

Jan 2025 – Present

- Used **multivariate time series models** with **LSTM** and **ARIMA**, to predict relative humidity, improving accuracy by 15% analyzing sensor responses and environmental variables, CO concentration.
- Performed data cleaning and **feature analysis** on 9358 hourly instances to identify key factors influencing RH.

Motto Systems Pvt. Ltd., Machine Learning Engineer Intern | Hyderabad, India

Sept 2022 – May 2023

- Engineered PySpark **ETL pipelines** using **AWS Glue** to process **10M+** pharma sales records, deployed **Prophet** models for 30-day demand forecasting (**15%** lower inventory costs), validated via **Athena SQL**.
- Implemented **k-means clustering model** to analyze 500K+ medical product transactions, identified 8 customer segments. Improved Click-through rate by **18%** with Bayesian-optimized regression and **Flask API**.
- Developed advanced **logistic regression** models in **Python** to stratify patient risk, rigorously validated with t-tests, and confirmed a **25%** risk reduction through targeted **A/B testing** of intervention strategies
- Automated **AWS Athena** queries to perform causal inference on promotional campaigns, applying Thompson sampling with PyMC3 for optimal budget allocation, which drove a **12% sales** uplift for 3 key therapeutics.

CSI Chapter, Technical Lead | Amaravati, India

Oct 2020 – Jul 2021

- Launched a **Django-based website** with **Python** and **SQLite** to display events and share workshop resources.
- Conducted **Power BI workshops** for 100+ students, hands-on training in data visualization with industrial support.

PROJECT EXPERIENCE

World Bank Debt Data Exploration SQL | PostgreSQL | Data Analysis

Jun 2024 – Aug 2024

- Analyzed World Bank international debt across 124 countries using **PostgreSQL** and **SQL** aggregations.
- Identified principal repayments as highest debt category with nested **SQL** queries for efficiency and ranked max country debts (\$96B) per indicator through optimized **SQL-based analytics**.

Wiki Comments Toxicity Identification | Natural language processing | BERT | Tensorflow

Apr 2024 – May 2024

- Developed Fine-tuned BERT on 63M Wiki comments, classifying toxicity across 6 categories, achieving 88.4% accuracy.
- Optimized macro-F1 threshold, increasing accuracy by 1.4% and effectively capturing nuanced toxic content.

Survival prediction system FastAPI | AWS | Data Analytics

Sept 2023 – Dec 2023

- Deployed a Titanic survival **prediction API** using FastAPI, enabling real-time predictions based on passenger features.
- Containerized the **FastAPI** application with **Docker**, ensuring scalable and reliable access to the model.

Festive-season Sales Performance Dashboard Power BI | SQL | Python

Jan 2023–Mar 2023

- Built **Power BI** dashboards using **SQL** to analyze Diwali sales, tracking **KPIs** like total sales, product categories, and sales by gender, age group, and occupation for real-time insights.
- Used **Python** and **Excel** to clean and process data, enhancing KPIs accuracy, improving sales forecasting by **20%**.

PUBLICATIONS

- Published author in Efficient system to predict harvest based on crop quality – Book chapter in [Springer](#).
- Published author in Efficient Wildlife Intrusion Detection System – [IEEE Xplore 2022](#).

HONORS

- Recipient of Dr. Gajwani Fellowship, based on excellence in academic records during my Masters tenure.